Sample Paper 11

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

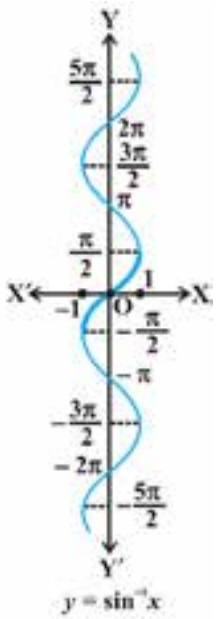
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the range of $y = \cos^{-1}x$ as represented in the graph?



(a) $[0,\pi]$ (b) [-1,1]

(c) $[-\pi/2,\pi/2]$ (d) $(-\infty,\infty)$

2. The solution of the differential equation $\frac{dy}{dx} + 1e^{x+y}$

(a) $(x+c)\,e^{x+y} = 0$ (b) $(x+y)\,e^{x+y} = 0$

(c) $(x-c)\,e^{x+y} = 1$ (d) $(x-c)\,e^{x+y} = 0$

3. Solution of $\frac{dy}{dx} + y\sec x = \tan x$ is

(a) $y(\sec x + \tan x) = \sec x + \tan x - x + c$

(b) $y = \sec x + \tan x - x + c$

(c) $y(\sec x + \tan x) = \sec x + \tan x + x + c$

(d) None of the above

4. The vector $\vec{a}$ is equal to

(a) $(\vec{a}\cdot\hat{k})\hat{i} + (\vec{a}\cdot\hat{i})\hat{j} + (\vec{a}\cdot\hat{j})\hat{k}$

(b) $(\vec{a}\cdot\hat{i})\hat{i} + (\vec{a}\cdot\hat{j})\hat{j} + (\vec{a}\cdot\hat{k})\hat{k}$

(c) $(\vec{a}\cdot\hat{j})\hat{i} + (\vec{a}\cdot\hat{k})\hat{j} + (\vec{a}\cdot\hat{i})\hat{k}$

(d) $(\vec{a}\cdot\vec{a})(\hat{i}+\hat{j}+\hat{k})$

5. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be vectors of magnitude 3, 4 and 5 respectively. If $\vec{a}$ is perpendicular to $(\vec{b}+\vec{c})$, $\vec{b}$ is perpendicular to $(\vec{c}+\vec{a})$ and $\vec{c}$ is perpendicular to $(\vec{a}+\vec{b})$, then the magnitude of the vector $\vec{a}+\vec{b}+\vec{c}$ is

(a) 5 (b) $5\sqrt{2}$

(c) $5\sqrt{3}$ (d) 2

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6. The least, value of the function $f(x) = ax + b/x$, $a > 0$, $b > 0$, $x > 0$ is

(a) $\sqrt{ab}$ (b) $2\sqrt{\frac{a}{b}}$

(c) $2\sqrt{\frac{b}{a}}$ (d) $2\sqrt{ab}$

7. A bag A contains 4 green and 3 red balls and bog B contains 4 red and 3 green balls. One bag is taken at random and a ball is drawn and noted to be green. The probability that it comes from bag B is

(a) $\frac{2}{7}$ (b) $\frac{2}{3}$

(c) $\frac{3}{7}$ (d) $\frac{1}{3}$

8. The equation of the line through the point $(2, 3, -5)$ and equally inclined to the axes are

(a) $x-2 = y-3 = z+5$

(b) $\frac{x-1}{2} = \frac{y-1}{3} = \frac{z-1}{-5}$

(c) $\frac{x}{2} = \frac{y}{3} = \frac{z}{-5}$

(d) none of these

9. Let $f(x) = x|x|$, then $f(x)$ has a

(a) local maxima at $x = 0$

(b) local minima at $x = 0$

(c) point of inflexion at $x = 0$

(d) none of the above

10. The value of $\int_{-2}^{2}(x\cos x + \sin x + 1)\,dx$ is

(a) 2 (b) 0

(c) -2 (d) 4

11. If x is measured in degrees, then $\frac{d}{dx}(\cos x)$ is equal to

(a) $-\sin x$ (b) $-\frac{180}{\pi}\sin x$

(c) $-\frac{\pi}{180}\sin x$ (d) $\sin x$

12. The image of the interval $[1, 3]$ under the mapping $f: R \to R$ given by $f(x) = 2x^3 - 24x + 107$ is

(a) $[75, 89]$ (b) $[74, 89]$

(c) $[0, 75]$ (d) none of these

13. $\int_0^{\pi/2}\left|\cos\left(\frac{x}{2}\right)\right|dx$ is equal to

(a) 1 (b) -2

(c) $\sqrt{2}$ (d) 0

14. The area of the region bounded by the lines $y = mx$, $x = 1$, $x = 2$ and X-axis is 6 sq units, then m is equal to

(a) 3 (b) 1

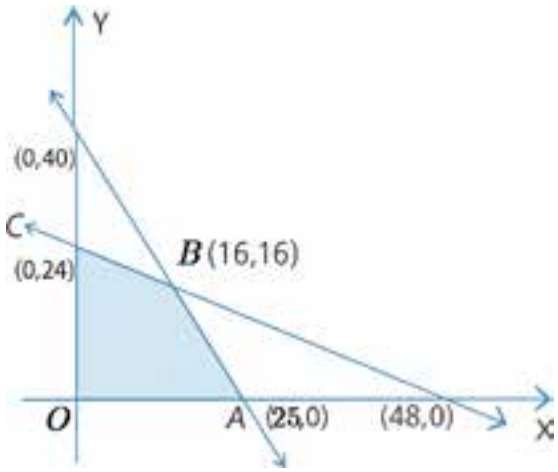
(c) 2 (d) 4

15. The value of $\sec^2(\tan^{-1}2) + \text{cosec}^2(\cot^{-1}3)$ is

(a) 15 (b) 5

(c) 13 (d) 14

16. Consider the objective functions $z_1 = 4x + 3y$ and $z_2 = 5x + 2y$. At which vertex of the feasible region do these two functions give the same value?



(a) A(25,0) (b) B(16,16)

(c) C(0,24) (d) None of these

17. In the figure, what does the element labeled "dx" represent?

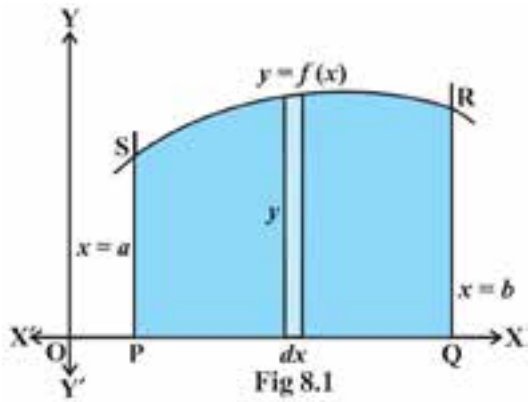


Fig 8.1

(a) The infinitesimal height of the strip parallel to the Y-axis

(b) The infinitesimal width of the strip along the X-axis

(c) The radius of the circle

(d) The area of the shaded region

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18. If matrix $\begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ \lambda & -3 & 0 \end{bmatrix}$ is singular, then λ is equal to

(a) -2 (b) -1

(c) 1 (d) 2

19. Consider the function $f(x) = \begin{cases} x^2, & x \geq 1 \\ x+1, & x < 1 \end{cases}$

Assertion: f is not derivable at $x = 1$ as $\lim\limits_{x \to 1^-} f(x) \neq \lim\limits_{x \to 1^+} f(x)$

Reason: If a function f is derivable at a point a then it is continuous at a

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

20. **Assertion:** Scalar matrix $A = [a_{ij}] = \begin{cases} k: & i = j \\ 0: & i \neq j \end{cases}$ where k is a scalar, in an identity matrix when $k = 1$

Reason: Every identity matrix is not a scalar matrix.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. A line passes through the point with position vector $2\hat{i} - \hat{j} + 4\hat{k}$ and is the direction of the vector $\hat{i} + \hat{j} - 2\hat{k}$. Find the equation of the line in cartesian form.

22. State the reason for the relation R in the set $\{1, 2, 3\}$ given by $R = \{(1, 2), (2, 1)\}$ not to be transitive.

23. Write the vector in the direction of the vector $\hat{i} - 2\hat{j} + 2\hat{k}$ that has magnitude 9 units.

OR

Find the magnitude of each of the two vectors $\vec{a}$ and $\vec{b}$, having the same magnitude such that the angle between them is 60° and their scalar product is $\frac{9}{2}$.

24. Evaluate $\int (1 - x)\sqrt{x}\, dx$.

OR

Given, $\int e^x (\tan x + 1) \sec x\, dx = e^x f(x) + C$. Write $f(x)$ satisfying above.

25. If $P(not\ A) = 0.7$, $P(B) = 0.7$ and $P(B/A) = 0.5$, then find $P(A/B)$.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. Find $\int \frac{dx}{5 - 8x - x^2}$.

27. If $\vec{a}$ and $\vec{b}$ are perpendicular vectors, $|\vec{a} + \vec{b}| = 13$ and $|\vec{a}| = 5$, then find the value of $|\vec{b}|$.

28. The volume of a cube is increasing at the rate of 8 cm^3/s. How fast is the surface area increasing when the length of its edge is 12 cm?

OR

The side of an equilateral triangle is increasing at the rate of 2 cm/s. At what rate is its area increasing, when the side of the triangle is 20 cm?

29. If $A = \begin{bmatrix} 5 & 6 & -3 \\ -4 & 3 & 2 \\ -4 & -7 & 3 \end{bmatrix}$, then write the cofactor of the element a_{21} of its 2nd row.

30. Write the value of $\sin\left[\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right]$.

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31. If $y = x\cos(a+y)$, then show that $\frac{dy}{dx} = \frac{\cos^2(a+y)}{\cos a}$
Also, show that $\frac{dy}{dx} = \cos a$, when $x = 0$.

OR

If $x = a\sec^3\theta$ and $y = a\tan^3\theta$, find $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{3}$.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. A manufacturer has three machine operators A, B and C. The first operator A produces 1% of defective items, whereas the other two operators B and C produces 5% and 7% defective items respectively. A is on the job for 50% of the time, B on the job 30% of the time and C on the job for 20% of the time. All the items are put into one stockpile and then one items is chosen at random from this and is found to be defective. What is the probability that it was produced by A?

OR

An insurance company insured 2000 scooter driver, 4000 car drivers and 6000 truck drivers. The probabilities of an accident for them are 0.01, 0.03 and 0.15 respectively. One of the insured persons meets with an accident. What is the probability that he is a scooter driver or a car driver?

33. Find the values of p, so that the lines

$$l_1 : \frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2} \text{ and}$$

$$l_2 : \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$$

are perpendicular to each other. Also, find the equation of a line passing through a point $(3, 2, -4)$ and parallel to line l_1.

34. Show that the differential equation $\frac{dy}{dx} = \frac{y^2}{xy - x^2}$ is homogeneous and also solve it.

35. Find $\int \frac{2\cos x}{(1-\sin x)(2-\cos^2 x)}\,dx$.

OR

Find $\int \frac{2\cos x}{(1-\sin x)(1+\sin^2 x)}\,dx$.

SECTION - E

Case study based questions are compulsory.

36. A craftswoman produces two products: floor lamps and table lamps. Production of one floor lamp requires 75 minutes of her labor and materials that cost \$25. Production of one table lamp requires 50 minutes of labor, and the materials cost \$20. The craftswoman wishes to work no more than 40 hours each week, and her financial resources allow her to pay no more than \$900 for materials each week.



(i) If she can sell as many lamps as she can make and if her profit is \$39 per floor lamp and \$33 per table lamp, how many floor lamps and how many table lamps should she make each week to maximize her weekly profit?

(ii) What is that maximum profit?

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37. The Indian toy industry is estimated to be worth US$1.5 billion, making up 0.5% of the global market share. The toy manufacturers in India can mostly be found in NCR, Mumbai, Karnataka, Tamil Nadu, and several smaller towns and cities across central states such as Chhattisgarh and Madhya Pradesh. The sector is fragmented with 90% of the market being unorganised. The toys industry has been predicted to grow to US$2-3 billion by 2024. The Indian toy industry only represents 0.5% of the global industry size indicating a large potential growth opportunity for Indian consumer product companies who will develop exciting innovations to deliver international quality standards at competitive prices.



Fisher Price is a leading toy manufacturer in India. Fisher Price produces x set per week at a total cost of $\frac{1}{25}x^2 + 3x + 100$. The produced quantity for his market is $x = 75 - 3p$ where p is the price set.

(i) Show that the maximum profit is obtained when about 30 toys are produced per week.

(ii) What is the price at maximum profit?

38. A manufacturing company has two service departments, S_1, S_2 and four production departments P_1, P_2, P_3 and P_4.

Overhead is allocated to the production departments for inclusion in the stock valuation. The analysis of benefits received by each department during the last quarter and the overhead expense incurred by each department were:

Service Department	**Percentages to be allocated to departments**					
	S_1	S_2	P_1	P_2	P_3	P_4
S_1	0	20	30	25	15	10
S_2	30	0	10	35	20	5
Direct overhead expense ₹'000	20	40	25	30	20	10



You are required to find out following using matrix method.

(i) Express the total overhead of the service departments in the form of simultaneous equations.

(ii) Express these equations in a matrix form and solve for total overhead of service departments using matrix inverse method.

(iii) Determine the total overhead to be allocated from each of S_1 and S_2 to the production department.

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